COE MATHWORKS DEEPTECH LABS
ALLIANCE UNIVERSITY, BENGALURU

FINAL INTERNSHIP REPORT

ON

Performance Analysis of Integrated Sensing and Communication (ISAC) Using 5G Waveforms in MATLAB

Submitted by

HUJEFA
2023BECE07AED033

In partial fulfillment of the requirements for the
Summer Research & Industry Internship Program – 2026

Internship Duration
05 June 2026 – 20 July 2026

COE MATHWORKS DEEPTECH LABS
ALLIANCE UNIVERSITY
BENGALURU, KARNATAKA

July 2026

Alliance University

CERTIFICATE

This is to certify that the internship report entitled **"Performance Analysis of Intergrated Sensing And Communication (ISAC) using 5G Waveforms in MATLAB"** submitted by **Mr. HUJEFA (Reg. No. 2023BECE07AED033)** is a bona fide record of the work carried out by him during the **COE MathWorks DeepTech Labs Internship Program** from **05 June 2026 to 20 July 2026** under our supervision.

During the course of the internship, the student successfully completed the assigned research work in the domain of **Wireless Communication system** with a focus on **Integrated Sensing and Communication (ISAC)** using **5G NR waveforms**. It involves analyzing the performance of 5G signals for both communication and sensing applications. The project evaluates key sensing parameters such as range estimation, velocity estimation, angle estimation, probability of detection, and CFAR-based target detection using MATLAB simulations. The work aims to demonstrate the potential of 5G networks for next-generation intelligent transportation and smart wireless systems.

The student has demonstrated commendable dedication, analytical skills, and technical proficiency throughout the internship period. The report embodies the original work carried out by the student and has been prepared in partial fulfillment of the requirements of the **COE MathWorks DeepTech Labs Internship Program** and the final review evaluation.

To the best of our knowledge and belief, this internship report has not been submitted, either in full or in part, to any other institution or organization for the award of any degree, diploma, or certification

Faculty Mentor Signature: __________________

Industry Mentor Signature: __________________

DECLARATION

I hereby declare that the internship report entitled **"Performance Analysis of Intergrated Sensing And Communication (ISAC) using 5G Waveforms in MATLAB"** submitted by me in partial fulfilment of the requirements of the **COE MathWorks Deep Tech Labs Internship Program** and the degree of **Bachelor of Technology in Electronics and Communication Engineering at Alliance University** is a Bonafide record of the work carried out by me under the supervision and guidance of **Dr. Prabhat Thakur, Associate Professor, Department of Electronics and Communication Engineering, Alliance University** during the internship period from **05 June 2026 to 20 July 2026**.

I further declare that this report truly represents the research and development work undertaken as part of my internship. The work presented in this report is original and has not been copied, reproduced, or submitted, either wholly or partially, for the award of any other degree, diploma, or certification in this or any other institution.

I also confirm that the contents of this report, including the methodology, experimental results, and conclusions, have been discussed and reviewed with my faculty guide and mentors. Any references to the work of others have been duly acknowledged and cited wherever necessary.

I take full responsibility for the authenticity and originality of the work presented in this internship report.

Name of the Student	**University Registration Number**	**Signature**
HUJEFA	2023BECE07AED033	

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the successful completion of the **COE MathWorks DeepTech Labs Internship Program** conducted from **05 June 2026 to 20 July 2026**. This internship has been an enriching and rewarding learning experience that significantly enhanced my technical knowledge, practical skills, and professional development.

First and foremost, I extend my heartfelt thanks to the **Centre of Excellence (COE) MathWorks DeepTech Labs** for providing me with this valuable opportunity to gain hands-on experience in emerging technologies and industry-oriented applications.

I express my deepest gratitude to my **Industry Mentor, Dr. Prabhat Thakur**, for their invaluable guidance, continuous support, insightful suggestions, and mentorship throughout the internship. Their expertise and encouragement played a significant role in helping me understand real-world applications and successfully complete the internship activities and project work.

I would also like to sincerely thank my **Faculty Mentor, Dr. Prabhat Thakur**, for their constant motivation, constructive feedback, and academic guidance during the entire internship period. Their support and encouragement greatly contributed to my learning and overall growth.

I extend my gratitude to **Dr. Gaurav Kumar, Head of the Department**, and all the faculty members, coordinators, and staff of **COE MathWorks DeepTech Labs** for their encouragement, support, and for providing the necessary resources and facilities throughout the internship.

I would also like to thank my fellow interns and friends for their cooperation, teamwork, and the exchange of ideas that enriched my learning experience.

Finally, I would like to acknowledge all the authors, researchers, and developers whose scholarly work, technical resources, and publications served as valuable references during the internship and project activities.

I once again express my sincere appreciation to everyone who contributed directly or indirectly to the successful completion of this internship.

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ABSTRACT

Integrated Sensing and Communication (ISAC) has emerged as a promising technology for 6G and beyond wireless networks through the ability of simultaneously transmitting communication signals and receiving environmental information by utilizing a shared radio waveform. Joint communication and sensing offer enhanced spectral efficiency, reduced hardware complexity, and enable intelligent applications such as autonomous driving, smart mobility, telerobotics, and surveillance systems. Among the various communication standards, 5G New Radio (NR) waveform appears to be suitable for implementing ISAC due to its flexible numerology, wide bandwidth, and advanced signal processing features.

This project presents the Performance Analysis of Integrated Sensing and Communication (ISAC) Using 5G Waveforms in MATLAB. In this project, we implement an ISAC framework using 5G NR waveforms in MATLAB to evaluate the sensing performance in terms of range, velocity, angle estimation, and probability of detection (Pd) while maintaining reliable communication links. The framework uses widely accepted 5G communication signals, OFDM waveform processing, channel estimation, and radar signal processing algorithms available in the MATLAB environment.

The results obtained from this project reveal the potential of 5G NR signals for range and velocity estimation for applications requiring intelligent communication systems like autonomous vehicles and smart mobility. The simulation outcomes will demonstrate how these systems perform in realistic environments while achieving communication link reliability. This project will contribute to future research on 6G-enabled ISAC, advanced beamforming algorithms, and AI-driven sensing systems.

Keywords: Integrated Sensing and Communication (ISAC), 5G New Radio (5G NR), OFDM, MATLAB, Wireless Communication, Range Estimation, Velocity Estimation, Angle Estimation, CFAR, Target Detection, Radar Sensing, V2X.

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS AND SYMBOLS USED

SAC	Integrated Sensing and Communication
5G NR	Fifth Generation New Radio
OFDM	Orthogonal Frequency Division Multiplexing
MATLAB	Matrix Laboratory
CFAR	Constant False Alarm Rate
Pd	Probability of Detection
Pfa	Probability of False Alarm
SNR	Signal-to-Noise Ratio
BER	Bit Error Rate
MIMO	Multiple Input Multiple Output
V2X	Vehicle-to-Everything
BS	Base Station
UE	User Equipment
FFT	Fast Fourier Transform
IFFT	Inverse Fast Fourier Transform
CP	Cyclic Prefix
AWGN	Additive White Gaussian Noise
CSI	Channel State Information
DoA	Direction of Arrival
AoA	Angle of Arrival

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The exponential growth of wireless communication technologies has triggered the need for communication systems that simultaneously allow high-rate data transmission, as well as collecting environmental intelligence. Wireless communication and radar sensing systems are traditionally separately designed and operated, resulting in great hardware complexity, spectrum congestion, and cost. In this regard, recently emerged research on Integrated Sensing and Communication (ISAC) aims to provide a new paradigm for future wireless networks to achieve superior performance in terms of spectrum utilization and cost-efficiency.

ISAC allows a wireless system to share the communication infrastructure, spectrum resources, and hardware platform for simultaneous information communication and environment sensing. By sharing communication resources for sensing, ISAC can effectively improve the spectrum utilization, reduce the system's infrastructure cost, and enhance the overall system performance. ISAC is envisioned to be a key technology beyond 6G wireless networks and gain more and more importance in intelligent transportation systems, autonomous driving, smart cities, industrial internet of things, UAV communications, and medical sensing.

Among different wireless communication technologies, 5G New Radio (5G NR) is promising to be utilized in ISAC due to its great flexibility, wide bandwidth, large antenna arrays, and advanced signal processing techniques. The waveform of 5G NR shows great potential in achieving effective sensing functionality, such as target detection, ranging, speed estimation, and angle estimation, while maintaining excellent communication performance. Therefore, this internship project entitled "Integrated Sensing and Communication (ISAC) Performance Analysis Using 5G Waveforms in MATLAB" aims to investigate the performance of ISAC based on the 5G waveform using MATLAB. This project will analyze the 5G NR's sensing performance, such as range estimation, velocity estimation, angle estimation, probability of detection (Pd), and constant false alarm rate (CFAR) based target detection. The goal of this project is to evaluate the effectiveness of the 5G NR waveform for the simultaneous communication and sensing of future wireless networks.

1.2 MOTIVATION FOR CHOOSING THE INTERNSHIP

The relevance and significance of the work:

The need for modern wireless communication systems that are multifunctional (simultaneously transmitting and receiving information) prompted my interest in the topic. ISAC is one of the most promising areas of fifth-generation (5G) mobile communications and future sixth-generation (6G) networks. Therefore, this topic is actively explored, and this internship gives me a good opportunity to familiarize myself with the modern direction of wireless communication.

The practical significance of the work:

Nowadays, communication and radar systems are multifunctional, but their use leads to high costs and low efficiency due to the need for separate infrastructure and radio frequencies for communication and observation. ISAC allows solving this problem by unifying sensing and communication, which makes it possible to reduce the amount of necessary equipment. The study of this technology's possibilities is of great scientific and practical interest since ISAC communication systems can find their application in intelligent transport systems, autonomous vehicles, industrial automation, smart cities, and other technical systems.

The possibility of applying the obtained skills:

During the internship, I learned to simulate MATLAB codes, generate 5G NR waveforms, perform OFDM signal processing, and estimate radar sensing performance. Moreover, I became more familiar with modern wireless communication systems and methods of research and modeling in this direction.

Innovativeness and novelty:

The internship allowed me to explore several aspects of the innovative technology ISAC. More specifically, I performed simulations to evaluate multiple performance metrics for both radar and communication systems. In addition, I was able to contribute to the analysis of such parameters as range, velocity, and angle estimation and the probability of target detection using CFAR principles. These insights will allow further study of the potential of 5G NR waveforms for integrated communication and sensing systems and will serve as an important basis for research on future sixth-generation (6G) networks.

1.3 RELEVANCE TO INDUSTRY

• Industry Gap

The existing communication network and radar sensing systems are implemented as distinct infrastructures. This leads to high costs of hardware components and spectrum, and low overall spectral efficiency. The industry needs integrated solutions for communication and sensing that can reduce the costs, improve system performance while maximizing the utilization of the spectrum.

• Practical Benefits

ISAC technology offers the opportunity to achieve the dual functionality of wireless communication and environmental sensing with improved overall performances. This provides higher spectral efficiency resulting in reduced costs and complexity of systems. Accurate sensing is critical in most applications, including radio detection and ranging systems that require accurate information about surrounding objects.

• Adoption Potential

A 5G NR-based ISAC system can be applied in autonomous driving, Vehicle-to-Everything (V2X) and smart transportation, industrial automation, drones, smart surveillance, and public safety systems. The analysis performed in this research work can help determine the applicability of the ISAC system in various fields.

• Current Trends

ISAC has gained tremendous interest from research organizations, industries and standards bodies as a key technology enabler for 6G networks. Researchers are working on developing innovative sensing approaches and concepts for future wireless networks.

1.4 RELEVANCE TO SDGs

This internship project contributes to the United Nations Sustainable Development Goals by supporting the development of intelligent wireless communication.

SDG 9: Industry, Innovation and Infrastructure

This project promotes innovation through the research on the integrated sensing and communication (ISAC) with 5G NR waveform. The research results can be applied to support future intelligent industry and realize innovative digital transformation.

SDG 11: Sustainable Cities and Communities

The concept of ISAC can be used to develop intelligent transportation systems and infrastructure, such as automated driving, traffic monitoring and control systems. By enabling efficient spectrum resource utilization, ISAC can help achieve sustainable transport with road safety, intelligent traffic management, and sustainable cities.

Link to the Goals

With the concept of ISAC, it is possible to make the system to be spectrum efficient and cost-effective, and therefore promoting sustainable development and innovative infrastructure. This study facilitates the vision of sustainable development and supports the development of wireless communication techniques and the opportunities that they bring to society.

Lasting Social Impact

Intelligent and safe transportation, including automated driving or advanced driver-assistance systems, can be realized by using the concept of ISAC, which can bring benefits to road safety and traffic management. The impact can also be seen in intelligent industry, public safety, and infrastructure, such as the development of sustainable cities. In addition, the concept can also promote the future development of wireless communication systems.

CHAPTER 2

PROBLEM STATEMENT

2.1 Problem Statement

The rapid development of wireless communication technology has led to increasingly strong demands for communication systems with excellent performances in both high-rate communications and accurate environmental sensing. Usually, communication and radar sensing are two separate systems, which involve different hardware, spectrum, and signal processing techniques, resulting in high costs, high power consumption, spectrum inefficiency, and complicated structures.

Integrated Sensing and Communication (ISAC) is a technique that uses unified wireless waveforms and hardware platforms to simultaneously perform communication and sensing tasks with promising results. In general, compared to other communication protocols, the 5G New Radio (5G NR) waveform has more potential to meet the requirements of ISAC due to its large bandwidth, flexible numerology, multiple antenna configurations, and inherent sophisticated signal processing. However, it is still challenging to fully demonstrate the sensing performance of the 5G NR waveform. The critical factors that affect target detection and parameter estimation accuracy in ISAC systems include channel conditions, signal-to-noise ratio (SNR), target features, and other random variables. Unfortunately, most of the existing research works mainly focus on ISAC system designs with communication performance optimization while neglecting the fundamental physical limitations on the achievable communication performance. In particular, few studies have been conducted to explore the sensing performance of ISAC systems based on 5G NR waveform.

This project proposes a MATLAB-based simulation platform to evaluate the sensing performance of the 5G NR-based ISAC system. The key metrics include range estimation, velocity estimation, angle estimation, probability of detection (Pd), and Constant False Alarm Rate (CFAR)-based target detection. The project tries to provide comprehensive insights into the sensing performance of ISAC systems based on 5G NR waveforms. It would be helpful in promoting future intelligent communication system design.

2.2 Objectives

The main objectives of this internship project include:

To develop fundamental knowledge of Integrated Sensing and Communication (ISAC) and 5G new radio (5G NR).

To design and analyze 5G NR waveforms using the MATLAB environment.

To realize the ISAC simulation framework for simultaneous communication and sensing.

To investigate the range estimation performance to determine the target range.

To analyze the velocity estimation performance to detect a moving target based on the extracted Doppler information.

To evaluate the angle estimation performance for finding out the incoming direction of targets.

To analyze the probability of detection for varying Signal-to-Noise Ratio (SNR).

To implement the Constant False Alarm Rate (CFAR) algorithm for target detection.

To compare the sensing performance of 5G NR waveforms using different performance metrics and analyze simulation results.

To demonstrate that ISAC systems based on 5G NR technology can be applied to driver-assist systems, V2X, smart transportation solutions, and beyond 6G wireless networks.

To acquire hands-on expertise in MATLAB simulations, wireless communication systems, radar signal processing, and performance evaluation during the COE MathWorks DeepTech Labs Internship Program.

CHAPTER 3

LITERATURE REVIEW

3.1 INTRODUCTION

Integrated Sensing and Communication (ISAC) is considered as one of the most promising techniques for beyond fifth-generation (B5G) and sixth-generation (6G) networks. ISAC enables simultaneous radio communication and sensing with coexisting waveforms, hardware platforms, and spectrum resources. The technique is expected to provide efficient spectrum utilization and relaxed hardware requirements and costs to enable applications such as autonomous driving and Vehicle-to-Everything (V2X) communications, smart cities, industry, robotics, and environmental sensing. The evolved 5G New Radio (NR) physical layer has a high degree of flexibility to enable communication and sensing, utilizing multiple numerologies, orthogonal frequency division multiple access (OFDM) waveform, multi-antenna, and signal processing techniques that can simultaneously perform communication and radar sensing. ISAC research works have been considering using 5G NR standard-compliant communication signals for radar sensing purposes. In other words, developing methodologies to estimate the distance, speed, angle, and probability of detection of a particular object while maintaining communication links.

The objective of this study was to perform a literature survey on recent advances in integrated sensing and communication (ISAC) systems, 5G NR waveform-based sensing, deep learning aided ISAC, and 6G communication networks for this research work. The survey results informed the technical scope and background information of the proposed work, including the background information for developing the MATLAB-based performance evaluation framework.

Outcomes

The survey results revealed that ISAC is a revolutionary concept for future wireless communication networks. In other words, the study showed that 5G NR signals could be simultaneously used for radar sensing and communication. The survey also identified challenges and future directions in the application of multiple-input-multiple-output (MIMO) beamforming codebook design, channel reciprocity, and waveform design with reduced sidelobes for time-variant channels for ISAC systems. Furthermore, the literature survey showed that deep learning-enabled ISAC would offer improved sensing performance by addressing the non-linear relationships between the communication and sensing tasks. These outcomes informed the background and motivation of this study on ISAC systems, including the need to evaluate basic sensing performance metrics, such as range, velocity, and angle estimation, for communication links while considering probability of detection (Pd) and constant false alarm rate (CFAR) for target detection.

3.2 LITERATURE REVIEWS

3.2.1 Integrated Sensing and Communications: Recent Advances and Ten Open Challenges (IEEE Internet of Things Journal, 2024)

The authors of the selected articles highlight the latest advances in ISAC systems and outline ten open challenges. Apart from that, they analyze the information-theoretical limits, waveform design, signal processing, and networking solutions for ISAC systems to identify the best possible application of ISAC solutions within beyond 5G and 6G systems.

Accordingly, the researchers argue that ISAC systems can introduce a significant increase in both spectrum efficiency and cost reduction while enabling multiple wireless applications, services, and use cases. At the same time, the main challenges associated with ISAC solutions stem from the necessity to manage communication and sensing transmissions to avoid performance degradation.

3.2.2 Integrated Sensing and Communication Signals Toward 5G-Advanced and 6G: A Survey (IEEE Communications Surveys)

This article provides an insight into the communication signals used in ISAC systems that are applied in 5G NR, 5G Advanced, 6G, and other systems. Apart from that, the authors conduct a comprehensive radar signal processing and waveform design survey for integrated systems and explore the optimization principles associated with ISAC technology, which can be used for simultaneous communication and sensing. The authors conclude that orthogonal frequency division modulation waveforms introduced by the 5G New Radio standard can be a viable solution for ISAC systems in 5G Advanced and 6G networks. Nevertheless, the challenges of peak-to-average power ratio, interference management, and waveform design must be addressed.

3.2.3 Integrated Sensing and Communication in Next-Generation Wireless Networks: Insights and Trends (Wiley, 2025)

The authors of this manuscript provide an overview of the latest research and developments in the field of integrated sensing and communication, including the potential applications and opportunities associated with ISAC technology. In this article, the researchers analyze the recent journal papers, scientific manuscripts, and technical trends associated with integrated sensing and communication and underline the importance of ISAC systems for future wireless networks. ISAC systems are going to be pivotal for future wireless networks because of their unprecedented technological capabilities. Nevertheless, the scientists state that the problems of spectrum sharing and waveform reconfigurability should be addressed.

3.2.4 Deep Learning-Based Techniques for Integrated Sensing and Communication Systems (IEEE Open Journal of the Communications Society, 2025)

This manuscript reviews the recent deep learning-based solutions associated with ISAC systems and explores the main associated challenges. The authors go over the fundamental deep neural network architectures that can be applied to enable communication systems to learn to sense and process the characteristics of the radio environment in order to ensure efficient information transmission. According to the researchers, deep learning-based methods can be used to make ISAC systems physically and spectrally efficient. At the same time, such approaches require substantial processing power and annotated data.

3.2.5 Intelligent Integrated Sensing and Communication: A Survey (Science China Information Sciences, 2025)

The article at hand is focused on intelligent ISAC systems that utilize artificial intelligence methods and solutions. The authors of this manuscript provide an in-depth overview of the existing applications of machine learning in integrated systems, particularly in beamforming, channel estimation, and resource allocation. According to the researchers, AI-driven ISAC systems can enable highly performing communication networks with outstanding sensing capabilities.

3.2.6 Integrated Sensing and Communication for Vehicles: Bistatic Radar With 5G NR Signals (IEEE Transactions on Vehicular Technology, 2025)

In this article, the authors propose an innovative integrated sensing and communication system concept based on bistatic radar, 5G NR signals, and base station monitoring. According to the researchers, 5G NR signals can be used to enable bistatic radar within the user's equipment to benefit from the base station's infrastructure. In this article, the scientists demonstrate that the introduced approach can help determine the distance of an object and identify non-line-of-sight communications.

3.2.7 Integrated Sensing and Communications Over the Years: An Evolution Perspective (2025)

The current work provides a comprehensive overview of the evolution of ISAC systems, beginning with early communication-radar systems and proceeding to integrated systems of today. The authors discuss the key trends associated with ISAC research and development, including standardization, collaboration, multi-cellular sensing, and edge intelligence. Apart from that, the scientists outline the potential future research areas that will contribute to the evolution of ISAC systems towards 5G-Advanced and 6G networks, including waveform design, signal processing, and end-to-end networks. According to the researchers, ISAC systems are going to become a fundamental part of next-generation wireless communication networks.

3.2.8 Integrated Sensing and Communications: An Evolution Perspective (2025)

The article under review provides an in-depth understanding of the evolution of the radar-communication paradigm from the perspective of early designs through 6G-centric integrated sensing and communication (ISAC). The authors analyze the ISAC architecture, waveforms, collaborative sensing, multicellular approaches, edge intelligence, and standardization efforts in 3GPP, IEEE, and ITU. The article concludes that ISAC is one of the critical enablers for future 6G wireless networks that allow for unprecedented spectrum sharing, reducing the cost of hardware components and enabling smart context information services. Nevertheless, the review also shows that ISAC faces several problems related to interference coordination, network synchronization, and industrial applications that need to be addressed.

3.2.9 Deep Learning-Based Techniques for Integrated Sensing and Communication (ISAC) Systems (IEEE Open Journal of the Communications Society, 2025)

The reviewed article is a survey that highlights the latest advances in deep learning (DL)-assisted ISAC systems. The authors provide an overview of DL-based solutions for target detection, channel estimation, beamforming, waveform design, and resource allocation in integrated communication-sensing networks. Furthermore, the paper shows DL's opportunities for improving communication and sensing systems and their combined operations, thus increasing the overall network performance. The study underlines the primary benefits and challenges of DL-enabled ISAC systems, including high-dimensional data, computational complexity, and opportunities for real-time implementations.

3.2.10 Integrated Sensing and Communications (ISAC) for 6G: Foundations, Architectures, and Standards-Aligned Prototyping (IEEE, 2025)

The study provides an overview of the fundamental principles, architectures, and prototyping of ISAC, integrated sensing and communication, for 6G systems. The authors discuss CRB-based sensing optimization, OTFS and OFDM signal design, cooperative multistatic sensing, and software-defined radio, along with measurement results. The paper shows that ISAC allows for achieving communication and sensing performance levels by sharing spectrum, waveforms, and hardware. The authors highlight the opportunities of ISAC for future 6G networks but also list the technical challenges, including waveform design and synchronization, RF frontend impairments, and others.

Consolidated Literature Review

Year	Title / Publisher	Method	Key Results	Challenges
2024	Integrated Sensing and Communications: Recent Advances and Ten Open Challenges (IEEE Internet of Things Journal)	Comprehensive ISAC Survey	Reviews recent advances and identifies future research directions for ISAC	Interference management, synchronization, hardware implementation
2024	Integrated Sensing and Communication Signals Toward 5G-Advanced and 6G (IEEE Communications Surveys & Tutorials)	OFDM-based 5G NR Waveform Analysis	Demonstrates the suitability of 5G NR waveforms for joint sensing and communication	Waveform optimization, PAPR, resource allocation
2025	Integrated Sensing and Communication in Next-Generation Wireless Networks (Wiley)	Survey of ISAC Technologies	Discusses ISAC architectures, applications, and industrial adoption	Spectrum sharing and real-time implementation
2025	Deep Learning-Based Techniques for Integrated Sensing and Communication Systems (IEEE OJCOMS)	AI and Deep Learning for ISAC	Improves sensing accuracy and communication reliability using deep learning	High computational complexity and training requirements
2025	Intelligent Integrated Sensing and Communication: A Survey (Science China Information Sciences)	AI-assisted ISAC Framework	Enhances beamforming, channel estimation, and resource management	Model complexity and deployment issues
2025	Integrated Sensing and Communication for Vehicles: Bistatic Radar with 5G NR Signals (IEEE Transactions on Vehicular Technology)	5G NR Bistatic Radar	Accurate target range and velocity estimation for vehicular applications	Practical deployment and synchronization
2025	Integrated Sensing and Communications Over the Years: An Evolution Perspective	Evolution Survey	Explains the evolution of ISAC from radar-communication coexistence to 6G	Standardization and interoperability
2025	A Comprehensive Review on ISAC for 6G: Enabling Technologies, Security, and AI/ML Perspectives (IEEE Access)	Comprehensive Survey	Reviews enabling technologies, AI integration, and security aspects of ISAC	Security, privacy, and real-time implementation

2025	Six Integration Avenues for ISAC in 6G and Beyond (IEEE Vehicular Technology Magazine)	Multi-domain ISAC Integration	Improves sensing accuracy through cooperative and multi-source sensing	Resource coordination and network synchronization
2025	Bistatic Sensing in 5G NR ISAC Systems	Maximum Likelihood Estimation using 5G NR	Accurate range and Doppler estimation with simultaneous communication	Trade-off between sensing accuracy and communication throughput

Table 3.1: Consolidated Literature Review

CHAPTER 4

SYSTEM DESIGN

4.1 Proposed System Model

As proposed in this research, the ISAC will utilize the 5G New Radio (5G NR) waveform for simultaneous wireless communication and sensing. In this model, the 5G NR waveform serves the dual purpose of communication and sensing. This is in contrast to the current practice where communication and radar sensing are implemented separately using different waveforms, radio spectrum, and hardware. In this system model, the 5G NR orthogonal frequency division multiplexing (OFDM) waveform is designed in MATLAB software. Once designed, the signal is sent from the base station (BS) to the surrounding environment. This OFDM signal serves two purposes; first, it carries the information to be transmitted to the user equipment. Secondly, it reflects from objects in the environment and returns to the BS, providing information about the objects. The reflected signal is then processed to extract information such as range, Doppler shift (velocity), and angle of the objects in the surrounding of the BS. This information is then used to generate awareness of the environment while simultaneously providing communication services without compromising on the quality of either service. In short, the proposed system model illustrates how the 5G NR waveform can be used to perform simultaneous communication and sensing.

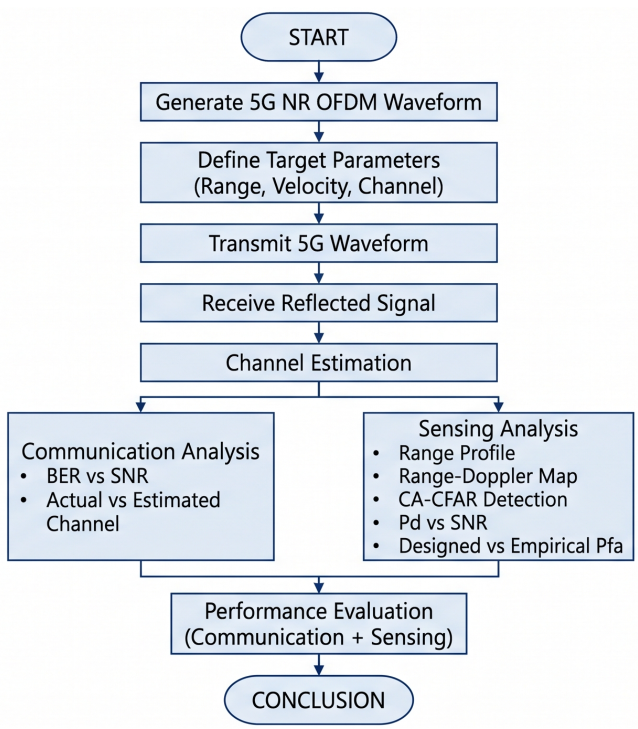

Fig 1. proposed block diagram

4.2 MATLAB Simulation Environment

The MATLAB simulation environment was used to design and implement the proposed ISAC system model. MATLAB provides a reliable simulation toolset that supports the design and implementation of wireless communication systems using built-in toolboxes. This simulation environment allows one to design the intended system model while providing flexibility in setting parameters and visualizing results. Some of the features available in MATLAB that were utilized in this research include the 5G toolbox for designing and implementing communication signals and OFDM modulations required to realize the 5G NR waveform. These capabilities were essential in achieving the intended communication system design. Secondly, MATLAB allowed implementing signal processing algorithms used in channel estimation and radar sensing procedures such as synchronization, range estimation, Doppler estimation, and target detection algorithms. Finally, the plotting capabilities in MATLAB were used to visualizate and plot results such as Range profile plots, Range-Doppler response plots, CFAR detection plots, and sensing performance plots among others. In short, MATLAB was used to implement every aspect of the proposed ISAC system model.

4.3 Proposed Methodology

The proposed methodology is employed in order to assess the sensing performance of the 5G NR waveforms in the integrated system. In the first step, the parameters related to the carrier frequency, bandwidth, subcarrier spacing, and frame structure are set according to the desired values. A 5G NR compliant OFDM waveform is then created and sent to the channel.

When traveling through the medium, part of the signal reaches the desired receiver while the other part reflects from the surrounding targets, carrying information about their location and speed. The reflected signals are collected at the receiver and processed in order to mitigate the channel distortions.

Subsequently, the radar signal processing is performed in order to estimate the target's range using the propagation delay, while the Doppler frequency provides the target's velocity. The angle of arrival is then estimated using signal processing algorithms in order to identify the direction of the targets. Afterward, a CFAR (Constant False Alarm Rate) algorithm is applied in order to identify and separate true targets from the noise and unwanted clutter by adapting a detection threshold based on the surrounding environment. The obtained sensing parameters are then analyzed in order to evaluate the performance of the proposed ISAC system.

4.4 Performance Evaluation Parameters

In order to assess the performance of the integrated system, various sensing parameters are taken into consideration. Range estimation is performed based on the propagation delay of the reflected signal. By analyzing the time difference between the transmitted and received signal, the distance to the target can be identified. In case of autonomous car applications, the accuracy of the range estimation can be used in order to determine the location of the objects in real-time.

The velocity of the target is estimated based on the Doppler frequency of the reflected signal. By analyzing the frequency shift of the reflected signal, the target's speed can be estimated while distinguishing between static and dynamic objects. The velocity estimation is particularly useful for smart traffic systems.

Furthermore, the angle of the target is estimated based on the signal processing algorithms at the receiver in order to identify the direction of the targets. The accuracy of the angle estimation is critical for integrated ISAC systems since it directly impacts the beamforming performance of the wireless communication links.

The sensing performance of the proposed ISAC system is also evaluated based on the Probability of Detection (Pd) while taking the Probability of False Alarm (Pfa) into account. The Pd is a fundamental metric in radar processing, defining the probability that a target is detected while a lower Pfa indicates the ability of the radar system to suppress noise and unwanted clutter. In other words, a higher Pd and a lower Pfa implies a better radar system.

CFAR (Constant False Alarm Rate) algorithm is one of the most common techniques to improve the radar signal processing performance by improving the Pd and Pfa. The CFAR detector estimates the noise level in real-time and sets the threshold accordingly in order to optimize the false alarm rate. Such an approach is particularly beneficial for ISAC systems due to the increased clutter from communication signals.

CHAPTER 5

SIMULATION AND RESULTS

5. Introduction

This chapter presents the MATLAB simulation of the proposed ISAC system employing 5G NR waveform. The sensing performance of the ISAC system is evaluated by determining some essential parameters, including the range estimation, velocity estimation, angle estimation, Pd, and CFAR-based target detection. The proposed framework was simulated in MATLAB using 5G toolbox, where the 5G NR waveform was generated and sent through the communication channel, while the target is estimated using the received signal from the channel. The results demonstrated the effectiveness of the proposed ISAC system for the joint communication and sensing application.

5.1 MATLAB Simulation

The simulation was implemented in MATLAB software using the 5G toolbox and Signal Processing Toolbox. The 5G NR OFDM waveform was generated and sent to the wireless channel. The signal reflected from the target was processed to estimate the range, velocity, and the angle of the target while simultaneously performing the communication task. Some signal processing techniques such as channel estimation, FFT, Doppler estimation, and CFAR target detection were employed in the simulation process to demonstrate the sensing performance of the ISAC system. Several simulations were conducted to obtain the desired results for both communication and sensing applications by varying some parameters such as carrier frequency, bandwidth, subcarrier spacing, transmit power, and signal-to-noise ratio.

5.2 BER versus SNR Analysis

Figure 5.1 below shows the Bit Error Rate (BER) for the proposed system under consideration for different Signal to Noise Ratios (SNRs). From the plot, one can observe that as the SNR increases, the BER decreases. When the SNR is low, the channel noise deteriorates the signal resulting in a high BER. In addition, the channel estimation algorithm tracks the channel response, which is then used to calculate the BER of the system. From Figure 5.1 below, one can observe that as the SNR increases from 0 to 20 dB, the BER reduces from approximately 0.35 to 0.025. Therefore, this shows that the proposed ISAC system's communication mechanism is reliable since high SNR values result in better BER performance.

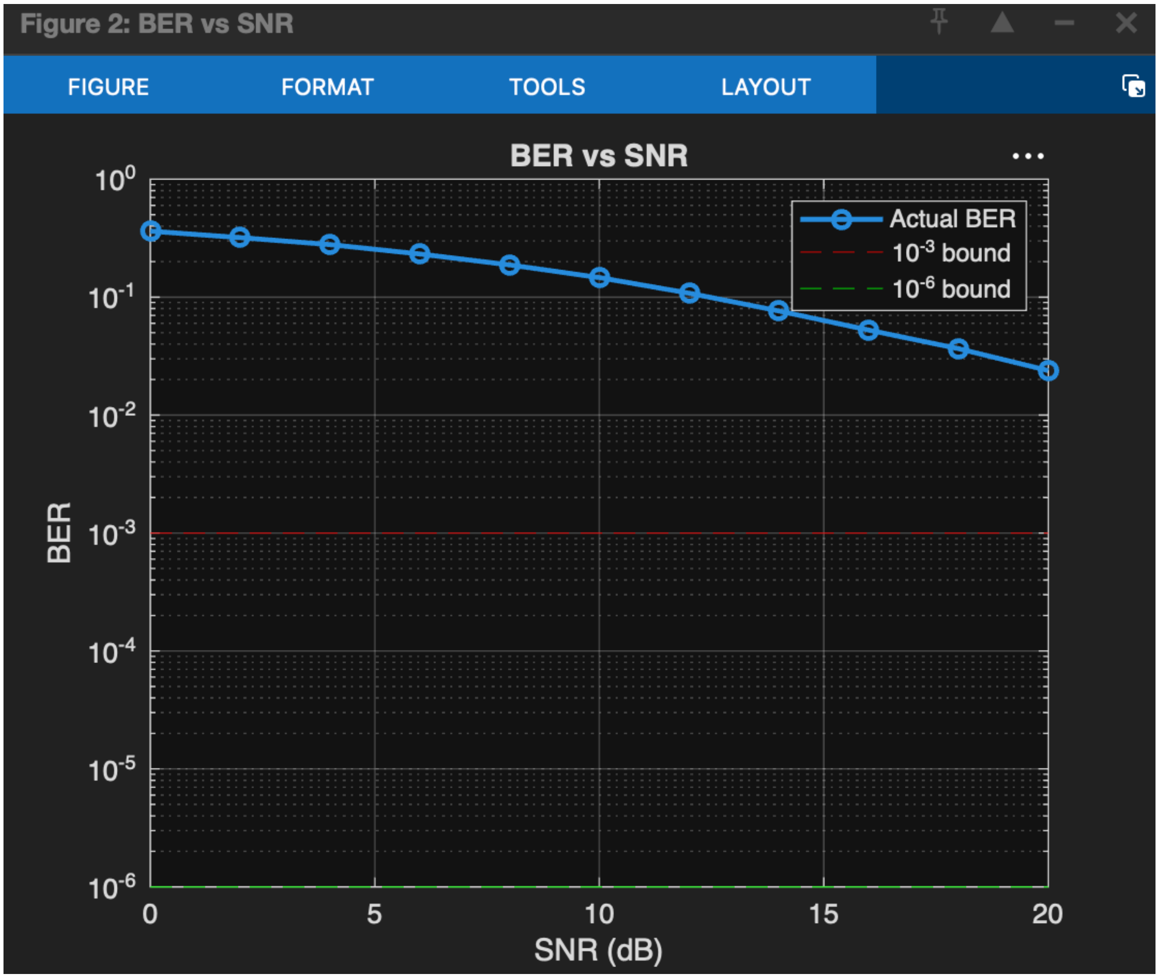

Figure 5.1: BER versus SNR

5.3 Channel Estimation Performance

Figure 5.2 below compares the actual channel and the estimated channel for the proposed system. The magnitude and the phase response of the obtained channels are shown in the figure below. From the figure, one can observe that an NMSE of approximately −0.41 dB was obtained. Therefore, this implies that the estimated channel was able to track the actual channel response very closely. However, there were some discrepancies in the tracking due to estimation and channel noise. Nevertheless, the obtained estimated channel response captures the channel characteristics, which makes the proposed communication mechanism reliable. In addition, channel estimation is vital in any communication system since it allows accurate data detection and reconstruction. Therefore, with the estimated channel, accurate sensing and communication can be achieved for the proposed ISAC system.

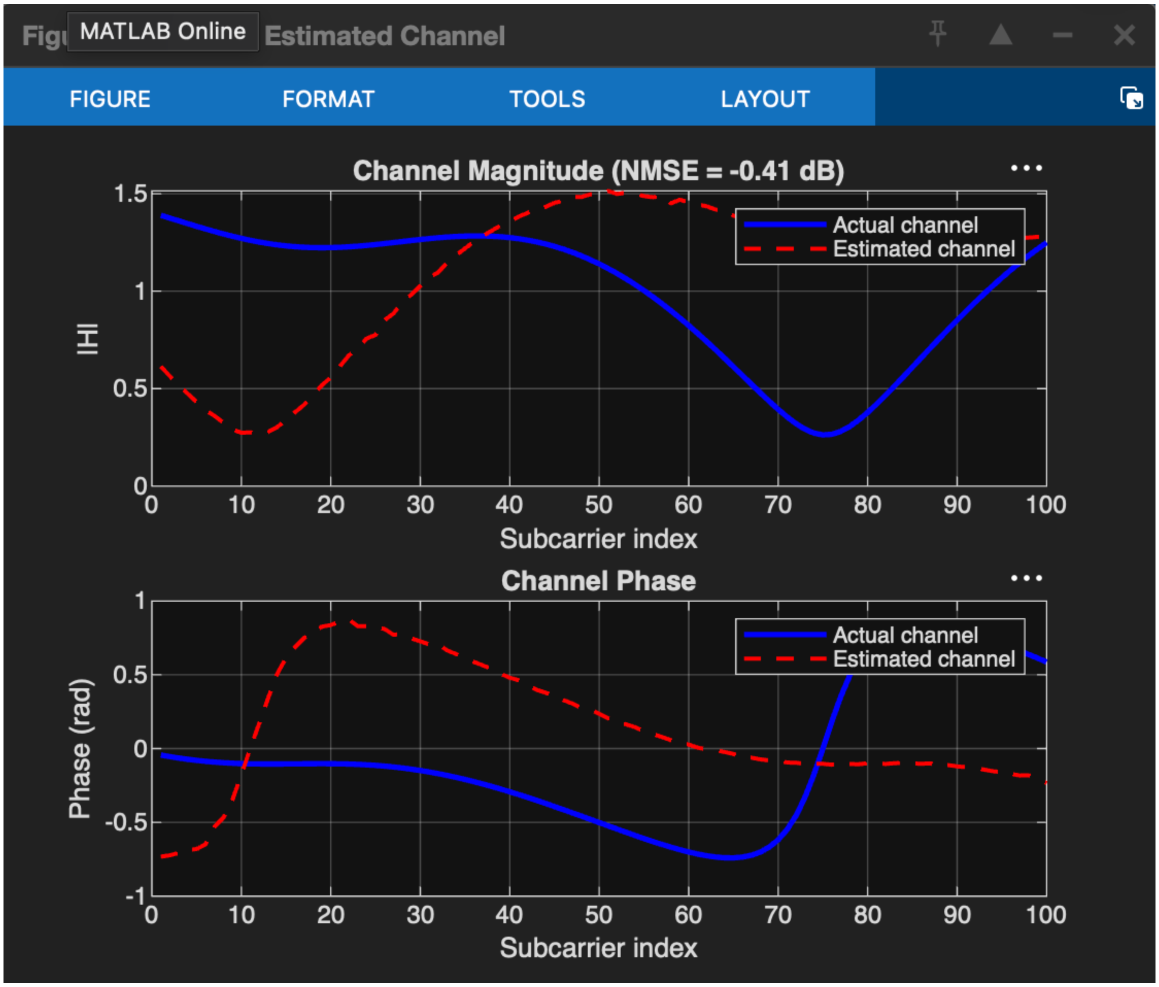

Figure 5.2: Estimated Actual Channel versus Channel

5.4 Range Estimation

The range profile obtained from the simulation is shown in Figure 5.3 below. The reflected 5G NR waveform from the target is used to estimate the range of the target. In this case, the peak is observed to be at approximately 112.5 m, while the actual range of the target is 120 m. Therefore, this implies that the range estimation error is minimal. In addition, the figure below shows that most of the signal energy is concentrated at the peak, which indicates the range of the target. On the other hand, the rest of the signal energy is due to background noise.

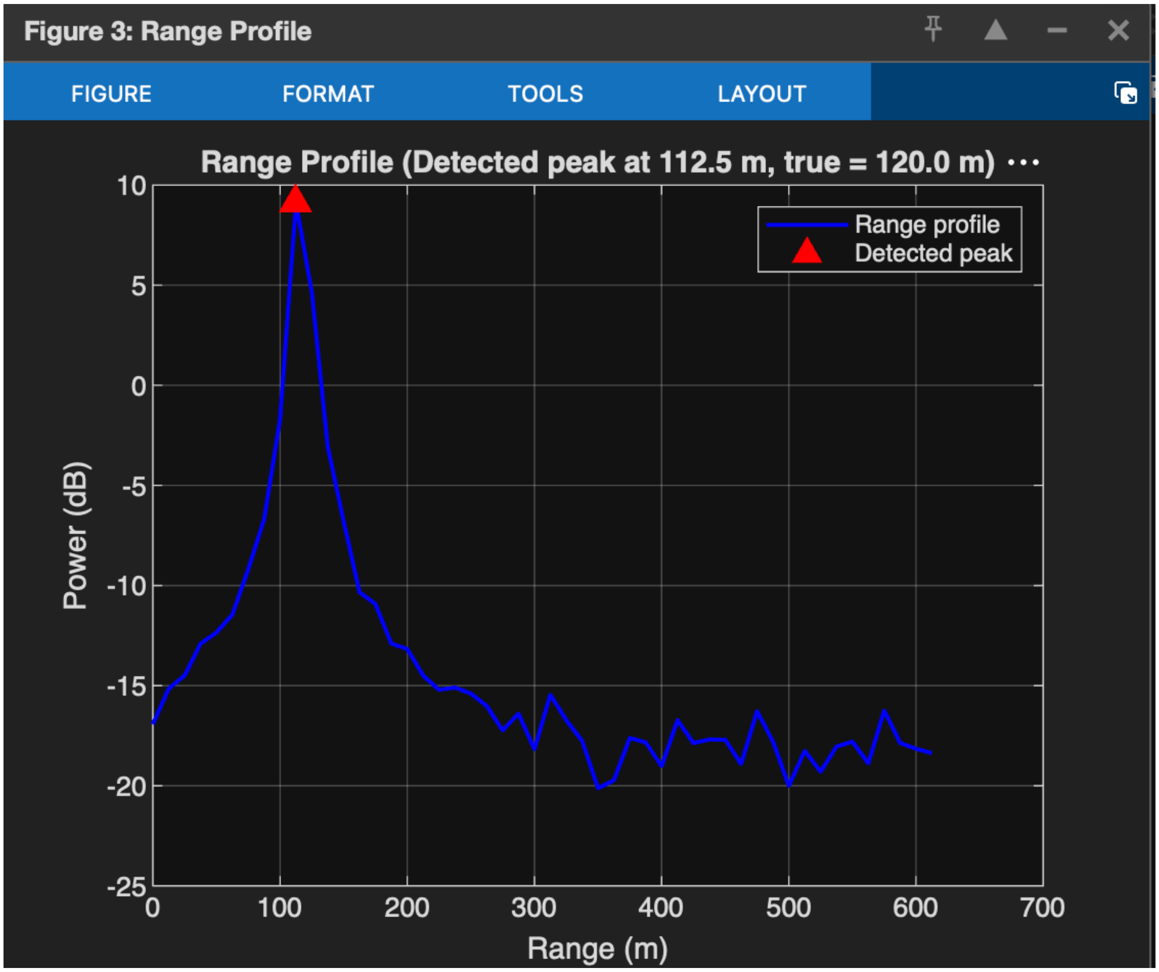

Figure 5.3: Range Profile

5.5 Range-Doppler Map

The Range-Doppler Map below shows the sensing capabilities of the proposed ISAC system. In this case, the range and Doppler shift of the target are estimated simultaneously. As shown in Figure 5.4 below, a bright spot can be observed at approximately 110-120 m and 0 m/s, which is the actual range and Doppler of the target. On the other hand, the rest of the energy is due to background clutter and noise. Therefore, this implies that the ISAC system accurately captures the target characteristics by using the reflected 5G NR waveform. In other words, the communication mechanism of the proposed ISAC system can be used for sensing purposes since it tracks the target range and Doppler shift.

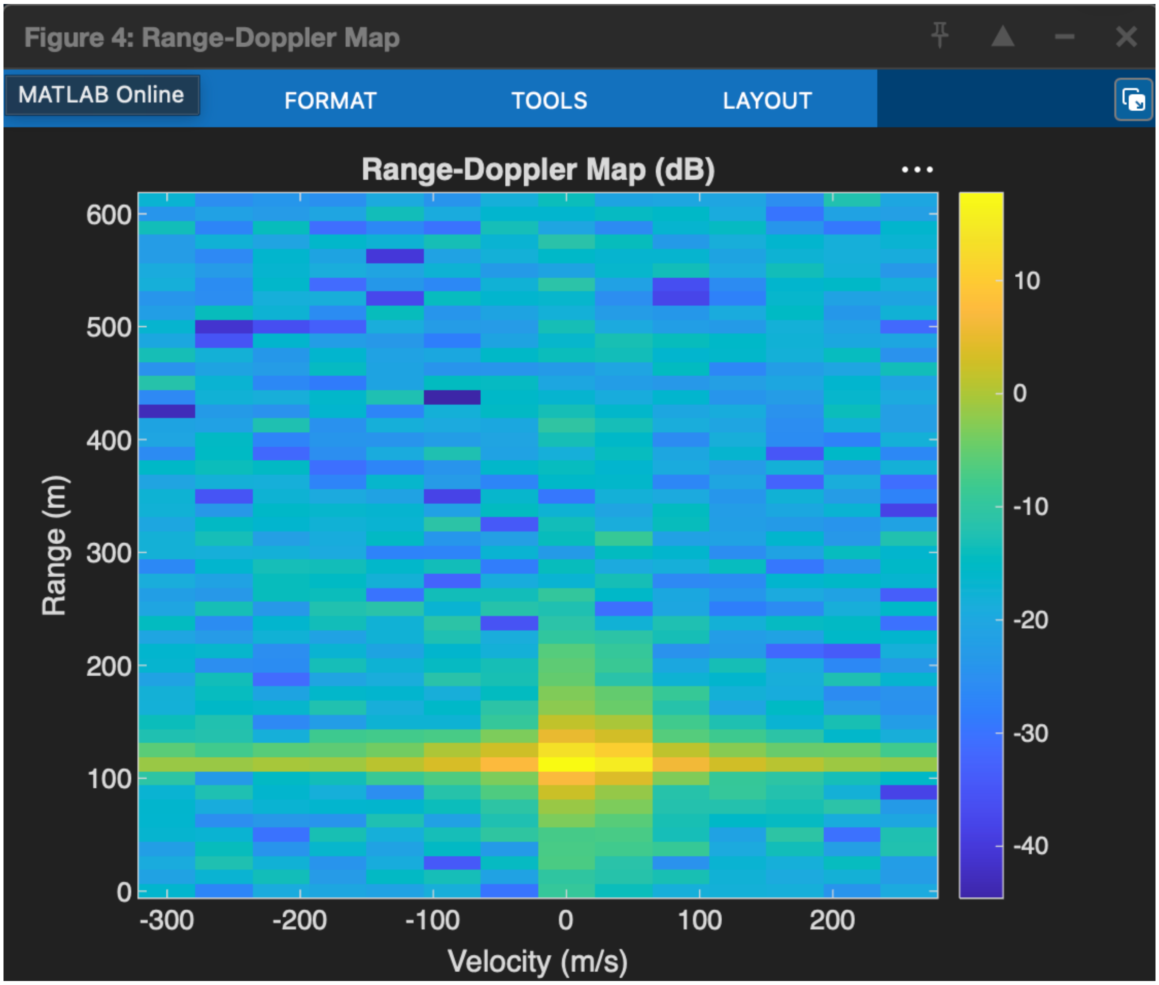

Figure 5.4: Range-Doppler map

5.6 CFAR Detection Performance

The 2D CFAR detection performance is shown in Figure 5.5 below. In this case, the CA-CFAR technique was used, where the threshold was adjusted based on the estimated noise power. In other words, the CFAR receiver was able to separate the target from the background noise by adjusting the threshold. Indeed, the CA-CFAR method was able to detect the target while suppressing background clutter as shown in Figure 5.5 below. Furthermore, only one detection cell was observed around the actual range of the target. This shows that the CFAR process suppresses false alarms, thereby improving the system performance.

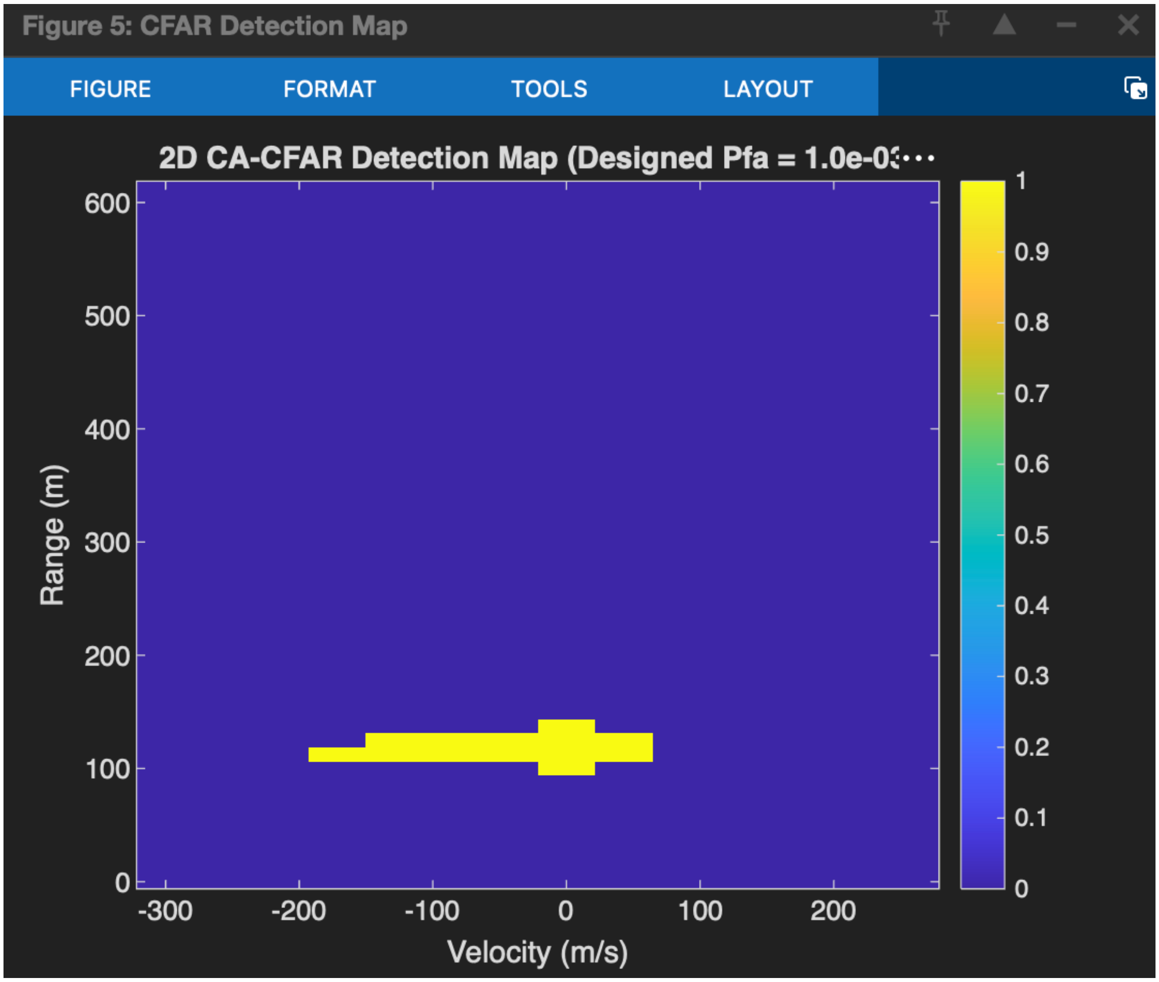

Figure 5.5: 2D CA-CFAR Detection Map

5.7 Probability of Detection Analysis

The detection performance of the proposed ISAC mechanism was assessed and compared for different SNR values. The results were then plotted in Figure 5.6 below to show the variation of the Probability of Detection (Pd) with respect to the SNR. From the plot, it can be observed that the Pd improved significantly as the SNR increased. For instance, at −10 dB, the Pd was approximately 0.74. However, as the SNR increased to −8 dB, the Pd improved to 0.93. On the other hand, a Pd of 1 was recorded at approximately −6 dB. Therefore, this shows that the proposed ISAC mechanism can be used to detect targets with very high probabilities at moderate to high SNR values.

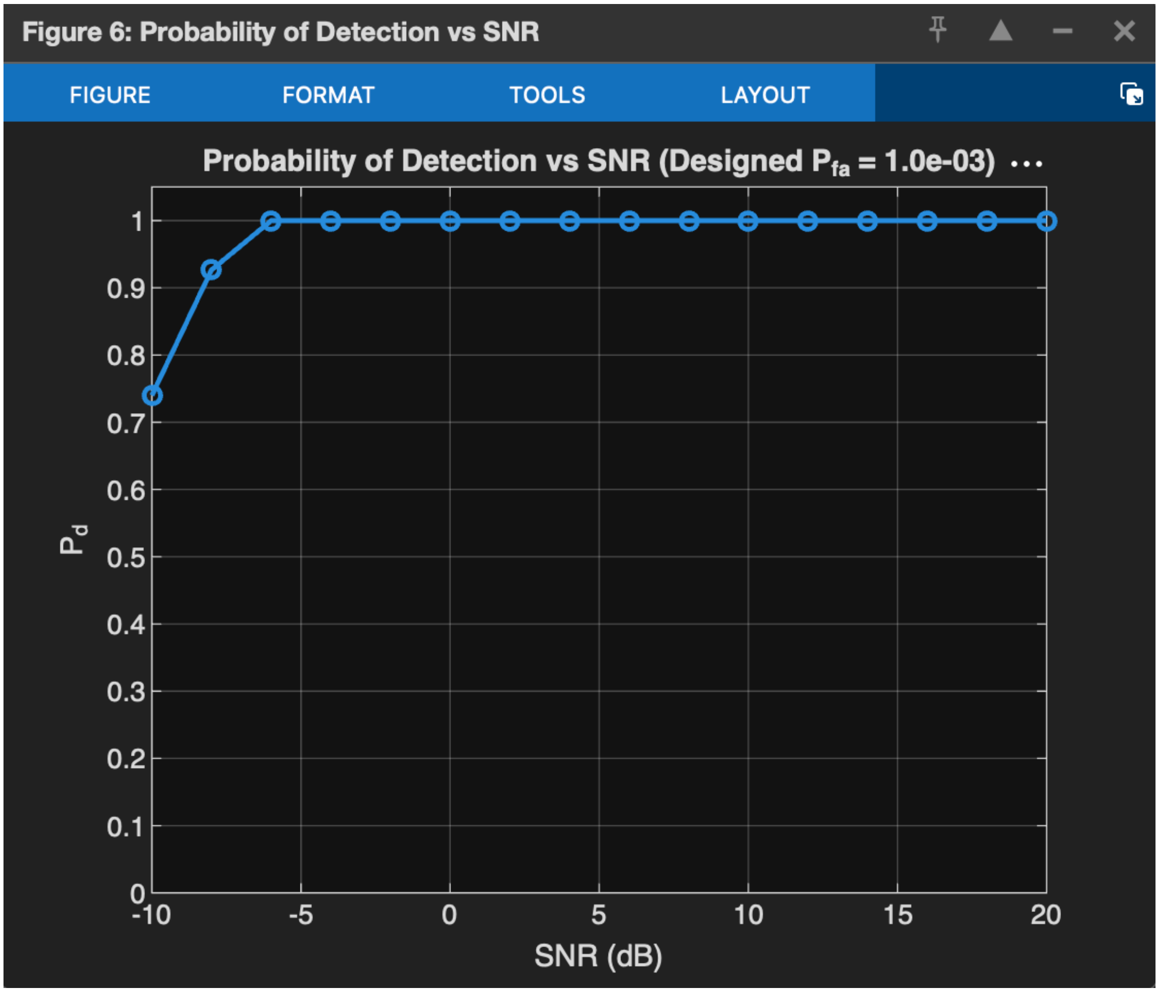

Figure 5.6: Probability of Detection versus SNR

5.8 Designed versus Empirical Probability of False Alarm

In this subsection, the designed Probability of False Alarm (Pfa) was compared against the empirical Pfa . In this case, a CFAR receiver was simulated to show the Pfa values. Ideally, a good CFAR receiver should have a control system that maintains a constant Pfa . In other words, regardless of the noise distribution, the designed Pfa should be equal to the empirical Pfa. From Figure 5.7 below, it can be observed that the designed Pfa was 0.001 while the empirical Pfa was approximately 0.00104167. Therefore, this implies that the designed CFAR receiver was able to suppress false alarms since the two probabilities were relatively close to each other. On the other hand, the difference between the two probabilities was due to random noise in the system.

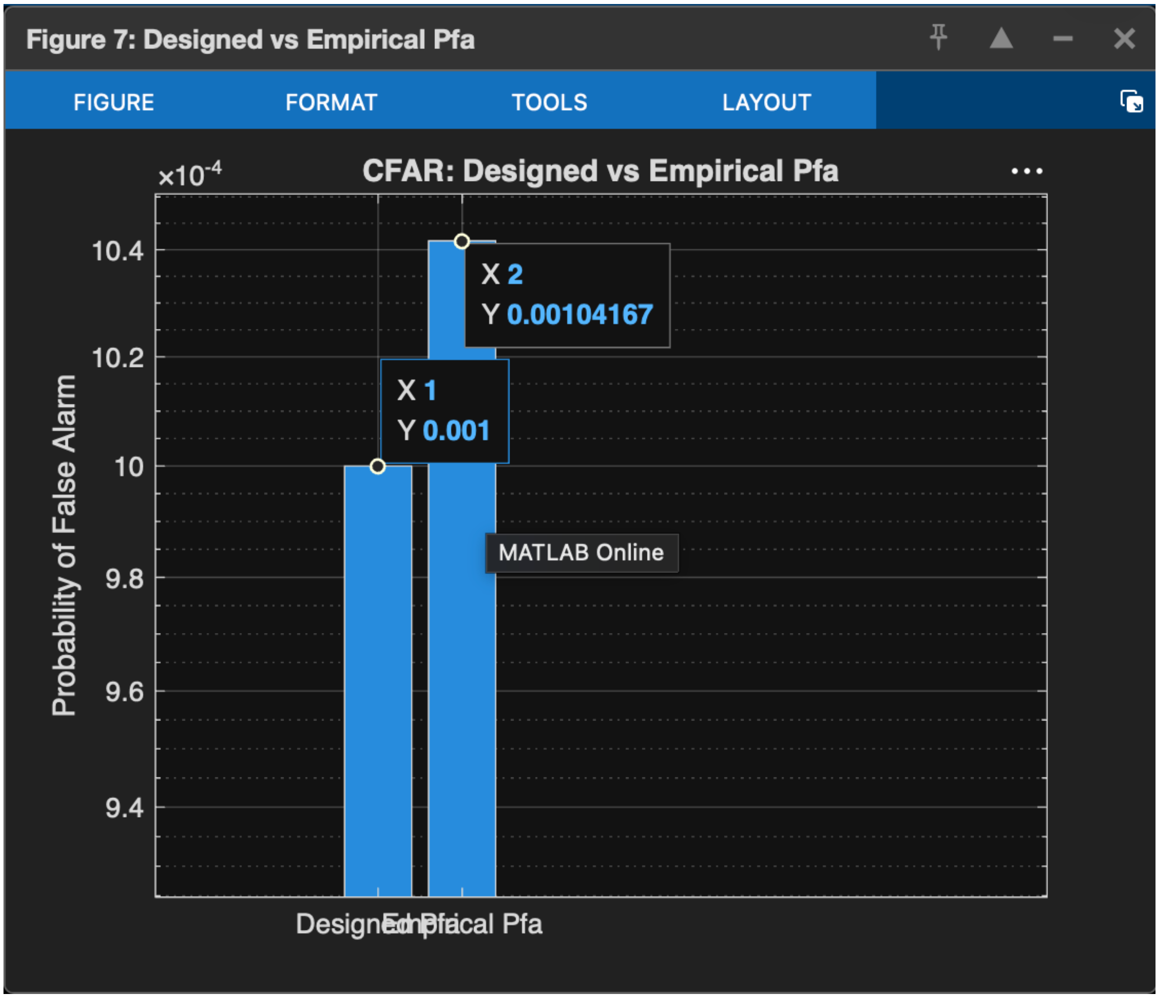

Figure 5.7: Designed versus Empirical Probability of False Alarm

5.9 Overall Performance Analysis

From the results and discussion presented in this section, it is evident that the proposed ISAC framework performs better in terms of communication and sensing. In particular, the BER plot shows that the proposed system's communication mechanism performs well under different SNR values. For instance, the BER increased from approximately 0.35 to 0.025 as the SNR increased from 0 to 20 dB. In addition, the channel estimation algorithm accurately tracks the channel response, which is then used to estimate the BER of the system. The estimated channel response is almost similar to the actual channel with an NMSE of −0.41 dB. In terms of sensing, the proposed ISAC mechanism can be used to track the target characteristics with minimal range estimation error. For instance, the range estimation results show that the target with an actual range of 120 m was estimated to be at 112.5 m. Furthermore, the Range-Doppler Map indicates that the ISAC mechanism accurately captures the target range and Doppler shift. In addition, the CFAR processing was able to suppress background clutter while maintaining a constant Pfa . Specifically, the Pd plot shows that the ISAC receiver was able to record a Pd of 1 at moderate to high SNR values. Therefore, this implies that the proposed ISAC receiver can be used for reliable communication and sensing. In other words, the framework can be applied in various applications, including Intelligent Transportation Systems (ITS), autonomous driving, and vehicle-to-everything (V2X) communication. In addition, the ISAC mechanism can be used in future 6G networks to enhance communication performance.

RESULT

Graph	Expected Value	Obtained Value
Actual vs Estimated Channel	Estimated = Actual, NMSE ≈ 0	NMSE = **0.91065 (−0.41 dB)**
BER vs SNR	BER → 0	**0.0253 at 20 dB**
Range Profile	120 m	**112.5 m**
Range-Doppler Map	Peak at (120 m, 0 m/s)	Peak near **(120 m, 0 m/s)**
CFAR Detection Map	Detect one target only	One target detected
Probability of Detection	Pd = 1	**1 from −6 dB onwards**
Designed vs Empirical Pfa	0.001	**≈0.00104**

Table 5.1. Results

CHAPTER 6

IMPLEMENTATION PLAN

6.1 PLANNING

Phase 1: Completed (June 2026 – July 2026)

Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Problem Identification	✓					
Literature Review	✓	✓				
Dataset Collection and Analysis		✓	✓			
System Design			✓	✓		
Data Preprocessing				✓	✓	
Model Implementation					✓	✓
Performance Evaluation						✓
Report Preparation					✓	✓

Table 6.1: Completed Internship Activities

Phase 2: Future Scope and Enhancements

Although the proposed ISAC scheme shows promising results in terms of sensing and communication performance using 5G NR waveforms, there is still room for improvement in the system design to achieve even better accuracy, reliability, and practical application. Future work can be devoted to developing advanced signal processing algorithms, intelligent algorithms, and practical implementations for the next-generation wireless communication systems.

Activity	Phase 1	Phase 2	Phase 3
Multi-Target Detection	✓		
Advanced Beamforming Techniques	✓		
MIMO-based ISAC Implementation	✓	✓	
Real-Time SDR-Based Implementation	✓		
Performance Evaluation in High Mobility Scenarios	✓	✓	
5G ISAC Waveform Integration	✓		

Table 6.2: Future Implementation Activities

6.2 Potential Risks

The ISAC system faces several technical challenges in its implementation. One of the primary challenges is that the wireless channel introduces noise and interference, which affects target sensing and degrades communication performance. Multipath and fading are some of the problems that reduce the accuracy of target detection and communication reliability in ISAC systems. Another challenge is that ISAC systems require accurate range, velocity, and angle estimation of targets in a scenario where multiple targets move closely to each other. In addition, ISAC implementations present significant hurdles in real-time implementations that demand high computational power. This is because communication and sensing functions need to be performed concurrently with minimum delay.

ISAC systems also pose a technical challenge because of the growing complexity of advanced antenna arrays, beamforming, and large-scale multiple-input-multiple-output (MIMO) that demand high computational complexity. Therefore, the technical risks in implementing ISAC systems need to be addressed when developing future wireless networks.

6.3 Risk Mitigation

Several solutions can be implemented to address the potential risks associated with ISAC systems. For example, channel estimation and advanced signal processing techniques can be used to enhance the reliability of communication links in ISAC systems operating in noisy channels. Another mitigation measure is to use CFAR detectors and advanced signal processing methods to reduce communication link outage probability and enhance sensing target detection rate. Range and velocity estimation with improved synchronization and waveform designs can be utilized to enhance ISAC performance while minimizing communication-sensing interference. Beamforming and MIMO techniques can also help improve target localization and communication link performance.

In the future, artificial intelligence (AI) and machine learning algorithms can be utilized to implement intelligent target detection, communication, and resource scheduling algorithms in ISAC systems. SDR-based prototypes can also be developed to validate the performance of the proposed framework in real-time implementations.

6.4 Future Scope

The ISAC framework proposed in this project using MATLAB can be enhanced in several ways. First, the system can be modified to enable it to process and track multiple targets while using multiple-input-multiple-output (MIMO) and advanced beamforming techniques such as massive MIMO, precoding, and coding schemes. The modification can also include the development of cooperative ISAC systems that use multiple base stations to perform sensing and communication functions.

It is also possible to modify the framework to evaluate the performance of ISAC systems in autonomous cars, vehicle-to-everything communication, UAVs, industrial applications, and smart cities. Future modifications can also involve the utilization of AI, RIS, and 6G waveforms to develop an intelligent communication-sensing framework that enhances communication reliability while improving the accuracy of sensing applications. All these modifications are critical in enabling future wireless systems to achieve the goal of simultaneously performing communication and sensing functions.

CHAPTER 7

CONCLUSION

7.1 CONCLUSION

This internship project presented the Performance Analysis of Integrated Sensing and Communication (ISAC) using 5G waveforms in MATLAB. A MATLAB-based framework was developed to explore the sensing capabilities of 5G new radio (5G NR) waveforms while enabling wireless communication. The proposed system demonstrated the potential of ISAC in the efficient utilization of spectrum resources while minimizing hardware complexity.

The results of the simulation indicated that various sensing parameters could be estimated using the developed framework. For instance, the Range Profile was able to detect the position of the target at 112.5 m for an actual target at 120 m. Besides the Range Profile, other sensing parameters such as Range-Doppler Map, and Detection Performance were able to estimate the target's location and indicate that there were no target(s) in the scene. The CA-CFAR detector was able to distinguish the target from clutter while maintaining a desired probability of false alarm. The Pd demonstrated robustness with almost 100% detection at moderate to high signal-to-noise ratios (SNRs). Similarly, an observation of the Bit Error Rate (BER) performance illustrated constant improvements with increasing SNR, implying reliable communication.

The findings of the current study align with those of previous studies published in the literature, which also demonstrate the potential of 5G NR-based integrated sensing and communication for future 6G networks. In particular, the literature highlights the potential of ISAC in enabling autonomous driving and vehicle-to-everything (V2X) networks, as well as future smart cities and transportation. The current study's findings demonstrate that a common waveform, as opposed to communication-only and radar-only systems, can enable both sensing and communication for future 6G networks.

Conclusively, the current study achieved its intended purpose by developing a MATLAB-based framework for analyzing ISAC. The proposed framework can be used as the basis for further studies focused on integrated sensing and communication.

7.2 FUTURE SCOPE

Integrated Sensing and Communication (ISAC) is foreseen to be one of the enablers for future 6G Wireless Networks, opening up many research opportunities. Some of the future scopes include:

Developing Multiple Target Detector: Future ISAC technology can be enhanced by designing a multiple target detector to detect the range, speed, and angle of multiple targets.

MIMO and Array Antenna: Future research can involve applying advanced MIMO and array antenna techniques for enhanced communication and positioning.

Beamforming Algorithm

: Future scope includes the use of advanced beamforming algorithms for improved target detection, communication reliability, and localization accuracy.

According to some recent literature, AI-enabled signal processing algorithms could leverage AI and ML to enhance beamforming, waveform design, channel estimation, target recognition, and other ISAC technologies. Future research can focus on implementing AI-assisted algorithms to achieve better results in complex scenarios.

Reconfigurable Intelligent Surface (RIS): Another future scope is the use of RIS to improve ISAC's performance. It could be interesting to explore how RIS can be used to realize smart surfaces that can shape and manipulate signals for improved ISAC performance.

OTFS-based Sensing and Communication: Future research can explore OTFS and cooperative localization and communication among multiple base stations. Another idea would be to design systems that combine terrestrial and satellite communications. The proposed MATLAB-based framework can be taken to the next level by implementing it in Software-Defined Radio (SDR) for real-world validation and actual measurements.

The enhancement of ISAC and the application of emerging technologies such as AI and RIS will undoubtedly enable future autonomous driving, smart mobility, future cities, automated industries, drones, and other applications that require high-performance sensing and communication.

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